When Medians Mislead: Revealing Hidden Temporal Structure in Child Crisis Disclosure

Boyoung Kang*
Sungkyunkwan University (SKKU)

Hayoung Oh†
Sungkyunkwan University (SKKU)

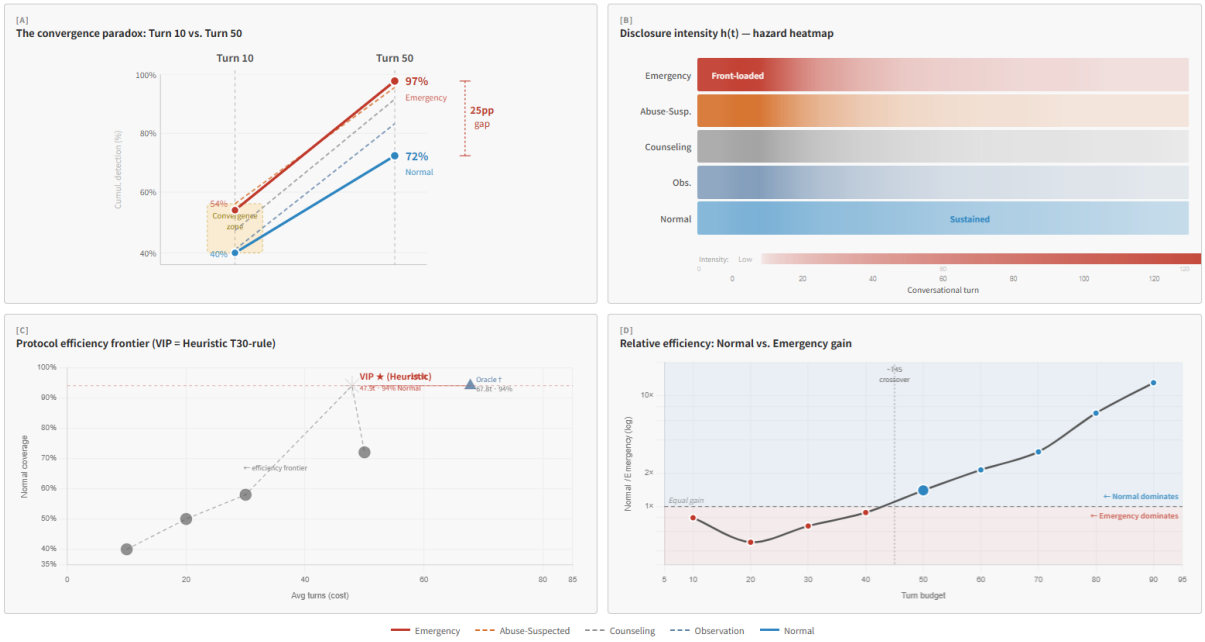

Figure 1: Four-view visualization of child crisis disclosure (N=3,236). Despite near-identical median timing (MSW_{50} = 9–11 turns), temporal trajectories reveal a 25pp detection gap at Turn 50: Emergency 97% vs. Normal 72%. **(C)** Protocol efficiency frontier: the Visually-Informed Protocol (VIP, also referred to as the Heuristic T30-rule) achieves 94% Normal coverage at 47.9 avg. turns — derived from the gain-ratio crossover at ∼Turn 45 visible in View D, which identifies Emergency saturation as the key decision boundary. A gain-ratio crossover at ∼Turn 45 (13× by Turn 90) justifies crisis-adaptive budgets reducing undetected Normal sessions from 42% to 6%. Aggregate statistics actively mislead; trajectory visualization makes the divergence observable — and actionable.

Abstract

Safety assessment in child counseling depends not only on *whether* risk is disclosed, but *when* it emerges. Current practice relies on aggregate timing statistics that collapse this temporal structure — and actively mislead. We present a four-view coordinated visualization of 3,236 Korean child counseling sessions that makes visible what summary statistics conceal. Aggregate metrics suggest near-identical disclosure timing across crisis levels (median 9–11 turns); our visualization reveals a 25-percentage-point divergence by mid-session: Emergency reaches 97% cumulative detection by Turn 50 while Normal reaches only 72%. This gap is invisible in any tabular summary — it becomes observable only through temporal trajectory visualization. Front-loaded Emergency cases versus slow, sustained Normal cases represent qualitatively distinct accumulation patterns unrecoverable from session-level statistics alone. The practical cost is concrete: uniform protocols leave up to 42% of Normal-context sessions undetected. A Visually-Informed Protocol (VIP) derived from trajectory visualization — requiring no prior knowledge of crisis level — reduces this to 6% at lower cost than oracle bounds, recovering 85.7% of the detection gap. When the structure of interest is temporal, aggregation is not simplification: it is distortion.

Keywords: temporal visualization, disclosure trajectories, child counseling, survival analysis, crisis detection, visual analytics, visually-informed protocol.

1 Introduction

Practitioners who assess child safety in counseling sessions face a decision at every turn: has enough risk information been gathered, or would continued interaction reveal critical signals? Current practice answers this question with aggregate statistics—and aggregate statistics give the wrong answer. Across 3,236 Korean child counseling sessions, median first-disclosure timing converges at 9–11 turns across all crisis severity levels. A practitioner or system designer reading this summary would conclude that a uniform interaction budget is sufficient. This conclusion is incorrect—and its cost is concrete. A protocol derived from this median alone leaves **42% of Normal-context sessions undetected**, systematically failing the children who disclose risk slowly rather than early.

The error is not recoverable from any tabular representation of the data. It requires visualization to become visible. When temporal trajectories are rendered as cumulative detection curves, a 25-percentage-point divergence emerges by mid-session between

*e-mail: boyoung.kang@skku.edu
†e-mail: hyoh79@skku.edu

Emergency (97%) and Normal (72%) cases—structure that is provably absent from median timing alone. We call this the *convergence paradox*: all crisis groups appear equivalent at Turn 10, masking divergence that only trajectory visualization can expose. This is not a presentational convenience. It is an instance of the broader principle that identical summary statistics can mask structurally distinct distributions [3, 22]: the median captures *when 50% of sessions disclose* but tells practitioners nothing about how the remaining 50% accumulate—information that directly governs whether to continue or conclude a session.

Prior work on counseling analysis [2,9,15], crisis detection [16,21, 23], and multi-turn dialogue research [8, 13, 18] consistently frames disclosure as a static, session-level classification outcome. This collapses conversational time into a single label, making temporal structure inaccessible by design. As conversational technologies assume increasing roles in mental health support [7, 10, 19], the inability to reason about *when* risk accumulates—and how hazard intensity varies across crisis levels—becomes an operational liability.

We address this gap with a four-view coordinated visualization designed to progressively expose what aggregate analysis conceals. Each view is motivated by a distinct analyst task: **View A** renders the convergence paradox directly—five crisis levels cluster within a 16-point band at Turn 10, then diverge 25 percentage points by Turn 50—making the gap immediately perceptible without calculation; **View B** reveals the qualitative mechanism underlying divergence—front-loaded versus sustained hazard intensity—unrecoverable from any tabular summary; **View C** translates visual structure into a protocol efficiency frontier, contrasting Uniform baselines against the Visually-Informed Protocol (VIP); **View D** provides the gain-ratio rationale for each budget boundary, where a crossover at ∼Turn 45 identifies Emergency saturation as the key decision point. Together, the views form a progressive sensemaking pipeline from paradox to mechanism to deployable decision rule.

This work makes three contributions: (1) we demonstrate that analysts relying on aggregate timing summaries systematically reach incorrect operational conclusions, and that temporal trajectory visualization is the minimal encoding that corrects this error; (2) we show that the convergence paradox arises from qualitatively distinct hazard structures—front-loaded versus sustained disclosure intensity—that are invisible in any static representation; and (3) we introduce VIP (Visually-Informed Protocol), a binary decision rule derived from trajectory visualization that requires no prior knowledge of crisis level, yet recovers 85.7% of the detection gap at lower cost than oracle bounds—reducing undetected Normal sessions from 42% to 6%.

2 Background and Motivation

2.1 The Temporal Decision Problem in Child Counseling

Child counseling requires continuous, turn-by-turn safety assessment under resource constraints. At each turn, the practitioner faces an implicit decision: has enough risk information been gathered, or would continued interaction yield clinically meaningful new signals? In current practice this judgment is guided by intuition alone, with no empirical basis for when to continue or conclude. As conversational technologies assume operational roles in mental health support [7, 10, 14], AI-mediated systems must operationalize session length as an explicit design parameter. The root problem is methodological: prior work treats risk disclosure as a *static* classification outcome [9], collapsing conversational time into a session-level label and making temporal structure inaccessible by design.

2.2 How Aggregate Statistics Induce Wrong Decisions

When practitioners seek empirical guidance, the answer appears straightforward: the median first-disclosure turn (MSW_{50}) is 9–11 turns across all five crisis levels in our corpus, implying that a uniform interaction budget suffices. The tools to detect the error — survival analysis, marginal gain computation — have long existed; what was missing was not the calculation, but the prompt to perform it. This conclusion is wrong. A Uniform-30 protocol achieves 87% Emergency coverage but only 58% Normal coverage— leaving 42% of Normal-context sessions undetected and representing a systematic safety gap in current protocol design. A Visually-Informed Protocol (VIP) derived from trajectory visualization closes this gap without prior knowledge of crisis level, recovering 85.7% of the detection gap and reducing the missed-session rate from 42% to 6%—at lower average cost than oracle bounds.

2.3 Why Visualization Is the Necessary Instrument

A median table produces the incorrect conclusion above. A classifier achieves near-ceiling accuracy on a trivially predictable label. A hazard table requires domain expertise and does not support visual comparison of accumulation dynamics. None enables a practitioner to *see* where trajectories diverge, *identify* group-specific saturation thresholds, or *estimate* a crisis-appropriate stopping point.

Rendering cumulative detection as visual trajectories supports three analytic tasks aggregate representations cannot:

- **T1 — Detect misleading convergence:** identify turns 9–11 where all groups appear equivalent, making the misleading median inference directly perceptible.
- **T2 — Locate divergence onset:** track the widening gap to 25pp by Turn 50 and reveal the qualitative mechanism—front-loaded versus sustained hazard intensity—without numerical subtraction.
- **T3 — Estimate stopping points:** identify group-specific saturation thresholds (i.e., where marginal gain per turn approaches zero) that operationalize crisis-adaptive turn-budget recommendations.

The 25pp gap is not hidden because it is uncomputable— it can be derived by subtracting two table entries. It is hidden because it is *visually occluded by aggregation*: no tabular representation makes the divergence perceptible, reveals its onset, or exposes the accumulation pattern that explains it. The numbers existed in Table 2; what did not exist was the recognition of Turn 30 as a strategic inflection point — that recognition required *seeing* the trajectory diverge and the gain ratio cross. We derive each visualization view from these analytic failures.

3 Methods

Design rationale. We derive four design requirements from the failure modes of aggregate analysis (Section 2): **(1)** Aggregate summaries suppress group differences by collapsing trajectories to single values; co-plotted multi-group curves with shared axes make accumulation dynamics simultaneously comparable without pairwise subtraction [5] (Views A–B). **(2)** A fixed-checkpoint snapshot reveals gap *magnitude* but not its *onset or growth*; an opacity-graduated fill encodes divergence as a perceptually growing spatial property, supporting pre-attentive perception without numeric estimation (View A). **(3)** Co-locating the MSW_{50} convergence zone and the 25pp divergence bracket within a single view forces the analyst to confront their inconsistency directly (Views A–C). **(4)** Absolute marginal gains compress to near-zero for both groups after Turn 30, obscuring their relative difference; a log-scale gain ratio (Normal / Emergency) makes the efficiency crossover at ∼Turn 45 directly perceptible, enabling VIP derivation without oracle assumptions (View D).

Dataset. We analyze the AI Hub Korean Child Counseling Corpus [1], a publicly available, fully de-identified dataset comprising 3,236 sessions (360,816 turns; ages 7–13; 2021–2023). Crisis levels were assigned by multidisciplinary clinical teams prior to any

conversational analysis and are used solely as external stratification variables.

Ethics and deployment stance. All data collection, consent, ethical review, and de-identification were completed by the data provider prior to public release. Our secondary analysis required no additional IRB approval [4]. Any positive safety signal should trigger referral to a licensed clinician; turn-budget metrics are decision-support instruments within human-supervised workflows, not replacements for clinical judgment.

Temporal anchoring. We identify the first child utterance containing any of 34 Korean risk keywords spanning eight semantic categories, developed with licensed child psychologists [6]. Detection uses exact substring matching on child turns, excluding negated and referential uses; sessions without matches are right-censored (8.7%). Robustness is confirmed via semantic similarity anchoring (cosine ≥ 0.50, [17]): median timing converges across paradigms ($\Delta < 1$ turn) with 5.2% additional coverage.

Survival and hazard analysis. Crisis-stratified cumulative detection $1-S(t)$ is estimated via Kaplan–Meier [11] with Greenwood 95% CIs; group differences are assessed via log-rank ($\chi^2 = 45.1$, $p < .001$). Turn-conditional intensity is modeled with discrete-time logistic hazard [20] using 10-turn bins and session-clustered SE (GEE). Marginal gains $\big([1-S(t)]-[1-S(t-10)]\big)$ are computed per crisis level and rendered as the Normal-to-Emergency ratio on a log scale, exposing the efficiency crossover that operationalizes View D. To assess operational utility, we conduct a counterfactual simulation comparing VIP and Crisis-Adaptive protocols against static Uniform-k baselines across all five crisis levels. Code available at the anonymous repository (`https://anonymous.4open.science/r/vis2026-crisis-disclosure-85EC`).

4 Results

4.1 Dataset and Temporal Anchoring

Our analysis covers 3,236 Korean child counseling sessions (360,816 turns; ages 7–13; Table 1). Sessions are approximately balanced across five crisis levels. Risk keywords were detected in 91.3% of sessions and used solely as temporal anchors. These keyword-based anchors prioritize transparency and auditability, with robustness validated via semantic similarity anchoring (see Robustness).

Physical Abuse keywords dominate (50–70%), with Emergency and Abuse-Suspected cases showing elevated Self-Harm and Social Issues co-occurrence ($\chi^2 > 100$, $p < .001$), supporting construct validity of the temporal anchoring lexicon (full category breakdown in supplementary materials). The MSW_{50} column makes groups appear equivalent — a conclusion Figure 2 directly refutes.

Table 1: Dataset and MSW_{50} by crisis level ($N = 3{,}236$)

Crisis Level	N	Turns M (SD)	Det.(%)	MSW_{50}
Emergency	672	123.5 (17.9)	92.4	9
Abuse-Suspected	644	117.9 (14.4)	94.4	9
Counseling	621	108.6 (15.3)	94.5	11
Observation	634	106.7 (13.5)	92.9	11
Normal	665	100.4 (13.2)	82.4	10
Total	3,236	111.5 (17.1)	91.3	–

Note. Det. = keyword-detected sessions. MSW_{50} values (9–11 turns) suggest parity across groups — a conclusion Table 2 and Figure 2 directly contradict.

4.2 The Central Paradox

The MSW_{50} column in Table 1 (9–11 turns across all groups) implies that a uniform interaction budget is sufficient. Table 2 contradicts this directly.

Table 2: Crisis-stratified cumulative detection at decision-relevant checkpoints

Crisis Level	N	MSW_{50} (turns)	T10 (%)	T20 (%)	T30 (%)	T50 (%)
Emergency	672	9	54	75	87	**97**
Abuse-Suspected	644	9	56	70	83	95
Counseling	621	11	47	66	78	91
Observation	634	11	41	55	66	83
Normal	665	10	40	50	58	**72**

Note. The 25pp gap between Emergency and Normal at Turn 50 is invisible in MSW_{50} alone. Log-rank: $\chi^2 = 45.1$, $p < .001$.

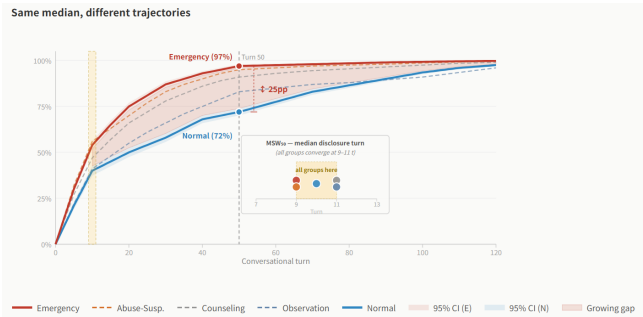

Figure 2: Temporal disclosure trajectories ($N = 3{,}236$, Kaplan–Meier). The growing red gradient visualizes the paradox: despite MSW_{50} convergence at 9–11 turns (inset), a 25pp gap emerges by Turn 50 — Emergency 97% vs. Normal 72%.

Figure 2 makes this structure immediately perceptible. The growing red gradient between Emergency and Normal trajectories encodes the widening gap as a spatial property: at Turn 10, all curves cluster near the MSW_{50} convergence zone (amber band, inset); by Turn 50, the gap is unmistakable. This difference cannot be inferred by subtracting rows in Table 2 — it becomes visible only when trajectories are rendered as a coordinated visual artifact.

4.3 What Aggregate Statistics Get Wrong: A Counterfactual Analysis

Table 3 quantifies the cost of relying on MSW_{50} alone: Normal coverage stalls at 72% even with Uniform-50, leaving a 28pp detection gap. The Visually-Informed Protocol (VIP) closes this gap without prior knowledge of crisis level — sessions with a keyword detected by Turn 30 terminate early; the remainder extend to Turn 100. This binary rule, derived directly from View D's efficiency crossover, achieves 94% Normal coverage at 47.9 average turns. Remarkably, VIP outperforms the oracle bound in efficiency (47.9 t vs. 67.8 t) while matching its coverage — instantiating Occam's Razor [12]: the simplest rule discoverable only through visualization outperforms the theoretically optimal one.

4.4 Why Trajectory Visualization: Encoding Comparison

Three representations of the same data reveal progressively more structure. A tabular MSW_{50} summary implies near-identical group behavior. A T50 bar chart reveals the 25pp endpoint gap but collapses the temporal process. Kaplan–Meier trajectories simultaneously expose convergence, divergence onset, and saturation dynamics — the three analytic tasks (T1–T3) no static representation supports. The numbers existed in Table 2; what did not exist was the recognition of Turn 30 as a strategic inflection point. Visualization is not the preferred medium for this finding — it is the condition for its discoverability.

Table 3: Protocol cost–benefit under aggregate vs. trajectory-based design ($N = 3{,}236$). Uniform protocols use MSW_{50} alone. VIP[§] (Visually-Informed Protocol) is a deployable binary rule derived from View D, requiring no prior knowledge of crisis level. Crisis-Adaptive[†] is an oracle upper bound with crisis level assumed known.

	Coverage by crisis level (%)						
Protocol	Emrg	Abus	Cnsl	Obs	Norm	Avg t	Undet.
Uniform-10	54	–	–	–	40	10 t	60%
Uniform-20	75	–	–	–	50	20 t	50%
Uniform-30	87	83	78	66	58	30 t	42%
Uniform-50	97	95	91	83	72	50 t	28%
VIP[§]	99	99	98	94	**94**	**47.9** t	**6%**
Crisis-Adaptive[†]	87	95	96	90	**94**	67.8 t	**6%**

Note. Coverage = cumulative detection at the protocol's turn budget (Table 2). Uniform-50 leaves 28% of Normal sessions undetected; both VIP and Crisis-Adaptive reduce this to 6%, recovering 85.7% of the gap toward perfect detection $\big((94-58)\,/\,(100-58)\big)$. VIP achieves identical Normal coverage to the oracle bound at 29.3% lower average turns (47.9 t vs. 67.8 t), requiring no prior knowledge of crisis level. [§]**VIP** (Visually-Informed Protocol): sessions with keyword detected by Turn 30 terminate at turn 30; remaining sessions extend to turn 100. Derived directly from View D's efficiency crossover. [†]Oracle upper bound: crisis level assumed known; budgets from Fig. 1D.

4.5 Mechanism and Crisis-Adaptive Implications

Hazard profiles reveal *why* trajectories diverge (Fig. 1B). All groups peak at turns 11–20 (HR = 1.196 vs. turns 1–10, $p < .001$), then decay. Emergency cases saturate rapidly after turn 30; Normal cases sustain 22% lower hazard through turn 100 (HR = 0.781 vs. Observation, $p < .001$). These front-loaded versus sustained structures are unrecoverable from session-level statistics and become observable only through the hazard view.

View D: Relative efficiency analysis. To provide an operational rationale for turn-budget boundaries, we render the per-turn gain ratio (Normal / Emergency) on a log scale (Fig. 1D). The visualization reveals a critical efficiency crossover at ∼Turn 45: prior to this point, Emergency cases yield higher per-turn discovery; beyond it, the relative utility of continuing Normal sessions increases monotonically, reaching 13× by Turn 90. This crossover directly generates VIP: terminate Emergency at ∼30 turns (saturation), extend Normal to ∼100 turns (sustained gain). The Crisis-Adaptive oracle instantiates the same insight with per-group precision, recovering 85.7% of the detection gap (Table 3).

4.6 Robustness

Semantic anchoring yields convergent median timing ($\Delta < 1$ turn) with 5.2% additional coverage; survival curves under three censoring scenarios are visually indistinguishable, with MSW_{50} identical across all conditions (range = 0 turns; supplementary materials). The visualized patterns reflect genuine disclosure dynamics, confirming that the representational critique in Section 2 targets a real structural property of the data.

5 Discussion

Visualization as a required analytical instrument. Prior analyses relying on aggregate statistics would conclude that disclosure timing is equivalent across crisis groups, leading directly to uniform interaction protocols. This conclusion is incorrect. Median disclosure timing (MSW_{50} = 9–11 turns) suggests parity; Kaplan–Meier trajectories reveal a 25pp detection gap that the median actively conceals — a concrete instance of what Anscombe [3] demonstrated. Temporal trajectory visualization is the representation required to make this structure observable, because no calculation would prompt a practitioner to look for it: it was the act of rendering the gain ratio that converted the hypothesis into VIP, a deployable rule requiring no prior knowledge of crisis level.

Design rationale. The four views form a progressive sense-making pipeline: View A renders the paradox; View B decomposes the mechanism; View C places protocols on an efficiency frontier; View D generates VIP via the gain-ratio crossover at ∼Turn 45. Neither VIP nor the oracle was learned by a model — both were designed by a practitioner reading the visualization, making the 85.7% gap recovery a human achievement enabled by visual analytics, not an algorithmic one.

Implications for visualization design in clinical NLP. The pipeline — temporal anchoring, survival modeling, coordinated visualization — is not specific to child counseling. Any multi-turn dialogue corpus annotated with outcome severity exhibits the same structural risk: aggregate summaries are computationally tractable but epistemically misleading. Visualization should be treated as a primary analytical instrument rather than a secondary communication layer, with temporal trajectory views accompanying — and potentially superseding — aggregate performance metrics in high-stakes conversational AI.

Limitations. Findings are grounded in Korean child counseling sessions (ages 7–13, 2021–2023) and require cross-cultural validation [10]. Keyword anchoring may miss metaphorical or implicit signals, and 8.7% of sessions without detected keywords remain an unresolved censoring challenge. A planned design study ($N = 8$–10 licensed counselors) will evaluate post-session reflection and decision revision.

6 Conclusion

Aggregate statistics can actively mislead. Across 3,236 Korean child counseling sessions, median disclosure timing (MSW_{50} = 9–11 turns) suggests functional equivalence across crisis groups — justifying uniform protocols. This conclusion is wrong. Temporal trajectories reveal a 25pp detection gap between Emergency (97%) and Normal (72%) at Turn 50, invisible in any tabular summary. VIP, derived from View D's gain-ratio crossover, reduces undetected Normal sessions from 42% to 6% at lower cost than oracle bounds. Temporal structure must be rendered, not summarized.

Three contributions of this work:

1. **A visualization paradox.** MSW_{50} convergence actively misleads protocol design; temporal trajectory visualization is the minimal encoding that makes the 25pp divergence observable.
2. **VIP: a deployable heuristic.** Derived from View D's crossover, VIP requires no prior knowledge of crisis level yet recovers 85.7% of the detection gap at lower cost than oracle bounds [12].
3. **A generalizable paradigm.** The temporal anchoring → survival modeling → coordinated visualization pipeline transfers to any multi-turn, temporally structured domain.

When the structure of interest is temporal, aggregation is not simplification — it is distortion. A single 30-turn checkpoint, derivable only by seeing the trajectory diverge, recovers 85.7% of the detection gap at no algorithmic cost. Crisis disclosure structure remains invisible under aggregate analysis, and becomes observable — and immediately deployable — only through time-aware visual analytics.

Acknowledgments

Generative AI tools (including Claude and ChatGPT) were used to assist with language refinement, code inspection, and figure preparation. All analyses, interpretations, and conclusions were performed solely by the authors.

References

[1] AI Hub. Psychological counseling data. https://aihub.or.kr/aihubdata/data/view.do?dataSetSn=71680, 2024. Accessed: January 2025. 2

[2] T. Althoff, K. Clark, and J. Leskovec. Large-scale analysis of counseling conversations: An application of nlp to mental health. *Transactions of the Association for Computational Linguistics*, 4:463–476, 2016. doi: 10.1162/tacl_a_00111 2

[3] F. J. Anscombe. Graphs in statistical analysis. *The american statistician*, 27(1):17–21, 1973. 2, 4

[4] Association for Computing Machinery. Acm policy on research involving human participants and subjects. https://www.acm.org/publications/policies/research-involving-human-participants-and-subjects, 2021. Accessed: 2026-04-12. 3

[5] S. Card, J. Mackinlay, and B. Shneiderman. *Readings in Information Visualization: Using Vision to Think*. Morgan Kaufmann, 1999. 2

[6] R. Chew, J. Bollenbacher, M. Wenger, J. Speer, and A. Kim. Llm-assisted content analysis: Using large language models to support deductive coding. *arXiv preprint arXiv:2306.14924*, 2023. doi: 10.48550/arXiv.2306.14924 3

[7] K. K. Fitzpatrick, A. Darcy, and M. Vierhile. Delivering cognitive behavior therapy to young adults with symptoms of depression and anxiety using a fully automated conversational agent (woebot): A randomized controlled trial. *JMIR Mental Health*, 4(2):e19, 2017. doi: 10.2196/mental.7785 2

[8] S.-H. Huang, Y.-F. Lin, Z. He, C.-Y. Huang, and T.-H. K. Huang. How does conversation length impact user's satisfaction? In *Proceedings of CHI '24*. ACM, 2024. doi: 10.1145/3613905.3650823 2

[9] Z. E. Imel, M. Steyvers, and D. C. Atkins. Computational psychotherapy research: Scaling up the evaluation of patient–provider interactions. *Psychotherapy*, 52(1):19–30, 2015. 2

[10] B. Kang and M. Hong. Development and evaluation of a mental health chatbot using chatgpt-4.0: Mixed methods user experience study with korean users. *JMIR Medical Informatics*, 13:e63538, 2025. doi: 10.2196/63538 2, 4

[11] E. L. Kaplan and P. Meier. Nonparametric estimation from incomplete observations. *Journal of the American Statistical Association*, 53(282):457–481, 1958. doi: 10.1080/01621459.1958.10501452 3

[12] P. King. William of ockham: Summa logicae. In *Central Works of Philosophy v1*, pp. 242–269. Routledge, 2015. 3, 4

[13] Y. Li, X. Shen, X. Yao, X. Ding, Y. Miao, R. Krishnan et al. Beyond single-turn: A survey on multi-turn interactions with large language models. *arXiv preprint arXiv:2504.04717*, 2025. doi: 10.48550/arXiv.2504.04717 2

[14] A. S. Miner, A. Milstein, and J. T. Hancock. Talking to machines about personal mental health problems. *JAMA*, 318(13):1217–1218, 2017. 2

[15] A. R. Pisani, M. S. Gould, C. Gallo, A. Ertefaie, C. Kelberman, D. Harrington et al. Individuals who text crisis text line: Key characteristics and opportunities for suicide prevention. *Suicide and Life-Threatening Behavior*, 52(6):567–582, 2022. doi: 10.1111/sltb.12872 2

[16] H. Qiu, L. Ma, and Z. Lan. PsyGUARD: An automated system for suicide detection and risk assessment in psychological counseling. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2024. 2

[17] N. Reimers and I. Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. In *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP)*, pp. 3982–3992, 2019. 3

[18] I. V. Serban, A. Sordoni, Y. Bengio, A. Courville, and J. Pineau. Building end-to-end dialogue systems using generative hierarchical neural network models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 30, 2016. 2

[19] S. Siddals, J. Torous, and A. Coxon. "it happened to be the perfect thing": Experiences of generative ai chatbots for mental health. *npj Mental Health Research*, 3(48), 2024. doi: 10.1038/s44184-024-00097-4 2

[20] K. Suresh, C. Severn, and D. Ghosh. Survival prediction models: an introduction to discrete-time modeling. *BMC Medical Research Methodology*, 22(1):207, 2022. doi: 10.1186/s12874-022-01679-6 3

[21] A. Swaminathan et al. Natural language processing system for rapid detection and intervention of mental health crisis chat messages. *npj Digital Medicine*, 6(1):213, 2023. doi: 10.1038/s41746-023-00951-3 2

[22] E. R. Tufte. *The Visual Display of Quantitative Information*. Graphics Press, 1983. 2

[23] T. Zhang, A. M. Schoene, S. Ji, and S. Ananiadou. Natural language processing applied to mental illness detection: A narrative review. *npj Digital Medicine*, 5(1):46, 2022. doi: 10.1038/s41746-022-00589-7 2

Supplementary Materials

Extended materials accompany this submission, including:

- **Appendix A:** Keyword category detection rates by crisis level and full co-occurrence heatmap.
- **Appendix B:** Censoring sensitivity analysis under three bounding scenarios with Kaplan–Meier overlays.
- **Appendix C:** Extended keyword taxonomy (34 terms $\times$ 8 categories, Korean + English).
- **Appendix D:** Full cumulative detection table at all checkpoints (T5–T100).
- **Appendix E:** GLM hazard model full coefficient table with session-clustered standard errors.
- **Appendix F:** Anonymous repository contents and quick-start instructions.
- **Appendix G:** Relative efficiency analysis — Normal/Emergency gain ratio table and log-scale figure (View D numerical basis).
- **Appendix H:** VIP simulation details — derivation of the T30-rule, weighted average turn computation, and full coverage breakdown across all five crisis levels.
- **Anonymous repository:** https://anonymous.4open.science/r/vis2026-crisis-disclosure-85EC — analysis code, keyword taxonomy, interactive HTML figure, gain ratio data, VIP simulation code, and synthetic data generator for full pipeline reproducibility.